Project Selection and Simple Corpus Analysis

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1 Project Overview

This report outlines our group selection and the initial stages of our corpus analysis project. Our focus is on character-level feature extraction for text classification, inspired by the architectures proposed by Zhang, Zhao, and LeCun (2015).

2 Character-Level CNN Architecture

The core approach involves treating text as a raw signal, processed through the following technical specifications:

Quantization	One-hot encoding using an alphabet of $m = 70$ characters (alphanumeric + punctuation).
Activation	ReLU activation: $h(x) = \max(0, x)$.
Pooling	Max-pooling layers to capture the most salient features across the sequence.
Optimization	SGD with momentum 0.9, starting $\eta = 0.01$ (halved every 3 epochs).
Minibatch	128 samples per iteration.

3 Proposed Project Components

1. **Introduction:** Defining the scope of character-level vs. word-level modeling.
2. **Related Work:** Reviewing Zhang et al. (2015) and existing CNN/RNN baselines.
3. **Initial Ideas:** Exploring the impact of character quantization on noisy datasets.
4. **Dataset/Corpus:** [Insert specific dataset name here, e.g., AG News or Yelp Reviews].
5. **Repository:** A well-organized GitHub structure for reproducibility.

4 Related Work Summary

> Zhang, Xiang, Junbo Zhao, and Yann LeCun. "Character-level convolutional networks for text classification." *Advances in Neural Information Processing Systems*, 2015.